

| Criteria                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Simple Definition                                                                                                                         | Indicators (Examples for more understanding)                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Review Level                                  | Dimension                                                                                         | Description                                                                                                                                                                          |
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| Unrepresentative Dataset Coverage                | The training or evaluation data for the feature does not adequately cover the learner groups, educational settings, or case types it is meant to serve. In educational technology, data dominated by one population or context can make aggregate results look acceptable while masking failures for underrepresented learner groups or settings. Flag PRs that add or change a model or dataset without visible coverage evidence, or where manifests, dataset schemas, sampling logic, coverage summaries, or evaluation outputs show missing intended groups, severe subgroup or setting imbalance, narrowly sourced benchmark data, or only aggregate results when subgroup labels are already available.                      | Data misses intended groups or settings.<br><br>If your data doesn't include everyone or every situation, your results can be misleading. | Flag PRs that change training, sampling, or eval data without subgroup coverage evidence, representative manifests, or group-level results.<br><br>Data mostly comes from one group, source, or setting<br>Some expected groups are missing or very small<br>Only overall results are shown (no breakdown by group)<br>No clear info about what the dataset covers                                                                                                                                        | Code level                                    | <b>Fairness-Evaluation</b><br>Fair & Representative Training<br>Data Assurance                    | Problems in data choice, group definitions, fairness targets, and thresholds used to judge fairness. It is about how fairness is measured before looking at later decision behavior. |
| Undefined Fairness Evaluation Attributes         | Fairness groups are unclear or too broad. Group definitions lack clear categories, granularity, or handling for missing values, making evaluation unreliable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fairness groups are unclear or too broad.<br><br>If you don't clearly name and track groups, you can't check if things are fair.          | Flag PRs that add or revise demographic fields without clear categories, subgroup granularity, missing-value handling, or a fallback evaluation plan.<br><br>No demographic or group fields are defined<br>Group categories are too broad (e.g., everything lumped together)<br>No explanation of what each group means<br>Missing or unclear handling of unknown/missing values<br>No plan to evaluate fairness if group data isn't collected                                                            | Code level                                    |                                                                                                   |                                                                                                                                                                                      |
| Missing Group-Wise Fairness Targets              | No fairness targets for each group. Models are assessed without defined group-level benchmarks, so disparities cannot be detected or enforced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | No fairness targets for each group<br><br>If you don't check each group separately, you might miss unfair results.                        | Flag PRs that validate models or rules with aggregate metrics only and omit group-level targets, assertions, or reports.<br><br>Only aggregate metrics are used (no group breakdowns)<br>No group-level metrics (e.g., accuracy per group)<br>No thresholds or targets for each group<br>No tests, alerts, or reports for group performance                                                                                                                                                               | Code level                                    |                                                                                                   |                                                                                                                                                                                      |
| Unvalidated Early-Warning Thresholds             | The system makes or surfaces early-stage predictions using thresholds, confidence rules, or missing-data defaults without visible evidence that those settings were validated for that stage of available data across learner groups. In educational technology, acting on predictions too early can create subgroup-skewed false positives or false negatives that affect support allocation, placement, or educator expectations. Flag PRs where release timing, threshold values, minimum-evidence requirements, or missing-data logic are added or changed without calibration results, subgroup error analysis, or tests showing acceptable performance before sufficient learner data is available.                          | Early-warning settings were not tested.<br><br>Don't use early guesses unless you've proven they work well.                               | Flag PRs that add early-warning thresholds, minimum-data rules, or missing-data defaults without subgroup calibration or timing validation.<br><br>Threshold values are set or changed without validation results<br>Predictions are made with very little data (too early stage)<br>No subgroup performance or error analysis is shown<br>Missing-data rules are used without testing impact<br>No tests or calibration for early-stage predictions                                                      | Code level                                    | <b>Pedagogy</b><br>Adaptive Fairness & Transparent<br>Learner Progression Integrity               | Problems where automation, predictions, or adaptation do not fit educational purpose or remove needed human judgment. It is about learning quality and proper educational use.       |
| Persistent Negative State                        | Negative scores, flags, penalties, or risk states persist across sessions or decisions without clear reset, decay, or re-evaluation logic, causing earlier errors or low-confidence judgments to shape later treatment of the same learner. In educational technology, this can escalate restrictions, interventions, or risk labels over time instead of reassessing each case on current evidence. Reviewable evidence includes cumulative scoring formulas, recursive flagging rules, history-based thresholds, state fields that carry forward adverse status, missing reset or decay conditions, and tests that fail to check whether repeated decisions amplify harm across cycles.                                          | Negative status keeps carrying forward.<br><br>The system doesn't "forget" or update negative judgments when it should.                   | Flag PRs that carry forward negative scores, flags, or penalties without reset, decay, re-evaluation logic, or anti-amplification tests.<br><br>Negative flags/scores are stored and reused across sessions<br>No reset, decay, or re-evaluation logic exists<br>Decisions depend heavily on past negative states<br>No checks to prevent repeated penalties over time<br>No tests ensuring old errors don't keep affecting new outcomes                                                                  | Code level                                    |                                                                                                   |                                                                                                                                                                                      |
| Hidden Learning Path Requirements                | Learning interfaces hide or obscure the requirements learners must satisfy to move through coursework. In educational technology, this causes avoidable failure when prerequisites, completion rules, locked-content reasons, deadlines, or required next actions are enforced in code or config but not clearly presented in the UI at the point of use. Reviewable failures include route guards or feature flags that block access without an explanatory state, labels or status text that do not identify what remains to be done, progress indicators that omit required steps, and missing tests for prerequisite visibility, locked-state messaging, and consistent next-step guidance across devices and assistive flows. | Steps to progress are hidden.<br><br>If requirements aren't clearly shown, learners can't move forward.                                   | Flag PRs that enforce prerequisites, locks, deadlines, or completion rules without visible explanations, next steps, or locked-state tests.<br><br>Access is blocked (locked content) without clear reason/message<br>Prerequisites exist in code but are not displayed in UI<br>No clear "what to do next" guidance<br>Progress indicators miss required steps<br>Missing or unclear messages for locked or incomplete states<br>No tests for showing requirements or next steps                         | Code level                                    |                                                                                                   |                                                                                                                                                                                      |
| Pedagogically Unconstrained Adaptation           | Adaptive system code keeps recommendations, sequencing, and interventions bounded by explicit pedagogical rules, course-design constraints, and learner-state checks rather than relying only on optimization scores. In educational technology, adaptation that ignores course intent or learner context can mis-sequence instruction, narrow options, and trap students in low-quality paths. Reviewable failures include recommenders or routing logic that act only on model scores or engagement metrics, missing prerequisite or progression constraints in rules or configuration, and threshold-based gating that lacks re-evaluation logic, override controls, or human-review checkpoints.                               | Adaptation ignores teaching rules or context.<br><br>The system adapts, but not in a smart or educationally correct way.                  | Flag PRs that let adaptive rules change sequencing or interventions using optimization scores alone without pedagogical constraints or override paths.<br><br>Recommendations or sequencing use only model scores or engagement metrics<br>No prerequisite or progression rules are enforced<br>Learner state (level, progress, needs) is ignored<br>No re-evaluation or update of decisions over time<br>No override, fallback, or human-review option<br>Threshold-based decisions lack safeguards      | Code level                                    |                                                                                                   |                                                                                                                                                                                      |
| Protected and Proxy-Based Differential Treatment | Protected attributes or close proxies, such as geography, language, or coded labels, are used in code, configuration, or decision logic to rank, score, route, gate, or recommend users differently in ways that can produce unequal treatment. In educational technology, this can change access, support, or outcomes for similar learners based on signals that stand in for protected status. Reviewable evidence includes protected or proxy fields in schemas, feature mappings, policy rules, filters, thresholds, weighting logic, and tests that apply different handling to comparable cases.                                                                                                                            | Protected or proxy traits change treatment.<br><br>Using personal traits (or hints of them) can lead to unfair treatment.                 | Flag PRs that use protected attributes or close proxies to score, route, rank, gate, or recommend users differently without an audited justification.<br><br>Protected attributes (e.g., race, gender) are used in logic<br>Proxy fields (e.g., location, language, ZIP code) influence outcomes<br>Different rules, thresholds, or weights are applied based on these fields<br>Similar users are treated differently due to these attributes<br>No justification or safeguards for using these features | Code level                                    | <b>Decision-Fairness</b><br>Protected Attribute Governance<br>& Validated Fairness<br>Enforcement | Problems in rules, model behavior, proxy use, protected-attribute handling, and fairness controls that change how learners are treated. It is about the decision path itself.        |
| Unvalidated Fairness Mitigations                 | Fairness fixes were not tested. Thresholds or rules are used without validation across groups or data conditions, risking biased or unstable outcomes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fairness fixes were not tested.<br><br>Problems are found, but no real or proven fix is applied.                                          | Flag PRs that add fairness mitigations without before/after subgroup metrics or regression tests showing the disparity improves.<br><br>Disparities are reported but no code/config changes follow<br>Fixes are added without before/after group metrics<br>No validation that the mitigation improves outcomes<br>No tests or checks for fairness after changes<br>Mitigation logic exists but lacks evidence or evaluation                                                                              | Code level<br>(Using previous review md file) |                                                                                                   |                                                                                                                                                                                      |

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| Unenforced Protected-Attribute Separation                            | Protected attributes are not given an explicit, enforced handling path that separates decision inputs from fairness-evaluation data. In educational technology, protected traits or close proxies can improperly influence predictions, rankings, alerts, or access decisions, while indiscriminate removal can also prevent reviewers from checking whether outcomes are equitable across groups. This issue is visible when code, schemas, feature definitions, prompts, config, data pipelines, or tests do not clearly enforce exclusion from decision logic, restricted retention for evaluation only, or a specifically justified and audited allowed use.                                                                                                                                         | Protected data is not kept separate.<br><br>You must separate sensitive data: don't use it for decisions, but keep it (safely) to check fairness.                                                       | Flag PRs that mix protected attributes into decision inputs or fail to enforce a separate evaluation-only path with restricted access.<br><br>Protected attributes (e.g., race, gender) are used directly in decisions<br>No clear rule separating decision inputs vs fairness-evaluation data<br>Protected data is removed entirely with no way to evaluate fairness<br>No access control or restriction on how these attributes are used<br>No documentation or justification for allowed use                                                                         | Code level |                                                                                       |                                                                                                                                                                                                    |
| Culturally Narrow Representations and Validation                     | Code, schema, model inputs, or configuration relies on culturally narrow representations, validation rules, or defaults that treat one naming, language, identity, or formatting pattern as normal and mis-handle users from other cultural contexts. In educational technology, this can prevent learners from entering valid information, force inaccurate self-representation, or degrade system behavior for users whose names, categories, or input formats do not match the assumed norm. Reviewable evidence includes ASCII-only validation, binary or limited identity fields, locale-specific assumptions for names, dates, currencies, or addresses, embedding or feature mappings that collapse distinct groups, and tests that cover only one cultural pattern.                              | Cultural defaults do not fit all users.<br><br>The system is built for one culture and doesn't handle others well.<br><br>The system assumes everyone is the same kind of user.                         | Flag PRs that assume ASCII-only names, binary-only identities, or locale-specific formats without inclusive alternatives and diverse tests.<br><br>Validation is too strict (e.g., ASCII-only names, limited characters)<br>Identity fields are too limited (e.g., only binary options)<br>Assumptions about formats (dates, names, addresses, currency) are fixed to one locale<br>Different groups are merged or simplified in data/embeddings<br>Tests only cover one cultural pattern                                                                               | Code level | Inclusion<br>Cultural & Accessibility Equity in Design and Code                       | Problems that exclude learners through interface language, validation rules, navigation, defaults, or resource assumptions. It is about practical access and participation for different learners. |
| Exclusionary Language in Code and UI                                 | Code, UI strings, error messages, comments, documentation, or identifiers use exclusionary, stigmatizing, or unnecessarily gendered language that can marginalize users or normalize harmful assumptions. In educational technology, this can make learners feel unwelcome, encode disrespect into everyday product interactions, and reinforce biased norms in systems used by diverse students and educators. Reviewable evidence includes deprecated discriminatory terms in identifiers, hostile or loaded wording in UI text or errors, gendered defaults where gender is irrelevant, and comments or documentation that describe some users as normal and others as deviations.                                                                                                                    | Language excludes or stigmatizes users.<br><br>The system uses words that can make people feel left out or disrespected.<br><br>Language in the system should be respectful and inclusive for everyone. | Flag PRs that add exclusionary, stigmatizing, deprecated, or unnecessarily gendered terms in identifiers, comments, errors, or UI strings.<br><br>Use of offensive, outdated, or stigmatizing terms<br>Gendered language where not needed (e.g., "he/she" instead of neutral terms)<br>Some users described as "normal" and others as different<br>Biased or disrespectful wording in UI text, errors, comments, or docs<br>Variable names or identifiers with problematic terms                                                                                        | Code level |                                                                                       |                                                                                                                                                                                                    |
| Excluding Access Assumptions                                         | Core user flows contain access assumptions that make the tool unusable for learners with disabilities or for learners using constrained devices, bandwidth, or locations. In educational technology, this directly blocks participation even when the underlying model or decision logic is otherwise fair. Reviewable failures include missing alt text or form labels, incorrect ARIA use, keyboard traps, color-only status signals, insufficient contrast, screen-reader-incompatible components, layouts that fail on mobile, large required assets with no low-bandwidth path, or configuration that blocks use by device or location without an equivalent educational path.                                                                                                                      | Access assumptions shut some learners out.<br><br>If you don't design for different needs and devices, some people get left out.                                                                        | Flag PRs that add learner-facing flows without labels, alt text, correct ARIA, keyboard access, contrast, mobile fit, or low-bandwidth paths.<br><br>Missing accessibility features (alt text, labels, keyboard support, ARIA)<br>UI relies only on color or has poor contrast<br>Not usable with screen readers<br>Layout breaks on mobile or small screens<br>Requires high bandwidth with no fallback<br>Blocks users by device, location, or connection without alternatives<br>wcag 2.1 aa standards                                                               | Code level |                                                                                       |                                                                                                                                                                                                    |
| Opaque AI Decision Outputs                                           | AI-driven decisions or recommendations are surfaced without clear, reviewable disclosures of what the output means, what information influenced it at a high level, and what confidence, uncertainty, or usage limits apply. In educational technology, students, educators, or administrators may over-trust or misread opaque scores, flags, placements, or recommendations when the system does not explain them well enough to interpret or challenge them. Reviewable failures include UI text, response schemas, audit or log fields, and tests or documentation that return only a prediction, label, score, or action without accompanying explanation content, confidence or context metadata, or plain-language statements about limitations and appropriate use.                              | AI outputs are not clearly explained.<br><br>AI should explain its decisions, not just give results.                                                                                                    | Flag PRs that expose AI scores, labels, placements, or recommendations without plain-language meaning, influencing factors, confidence, or limits.<br><br>Only a score, label, or decision is returned<br>No explanation of what influenced the result<br>No confidence or uncertainty information<br>No description of limitations or proper use<br>Missing fields for explanation, metadata, or logs                                                                                                                                                                  | Code level | Transparency<br>Transparent, Explainable, Validated & Human-Supervised AI Decisioning | Problems in explanations, labels, analytics wording, and result framing that make system behavior hard to understand or easy to misread. It is about correct interpretation of outputs and claims. |
| Causal Overclaiming in Analytics                                     | Analytics dashboards, reports, or generated insights describe observational patterns as causes or explanations of student outcomes without clear qualification. In educational technology, causal overclaiming can mislead educators into acting on attendance, behavior, engagement, or performance associations as though the system has shown why a result occurred. Reviewable failures include UI copy, chart titles, labels, annotations, metric definitions, tooltip or help text, generated summaries, and tests or documentation that frame correlational measures as causal findings instead of explicitly limiting them to association, trend, or risk information.                                                                                                                           | Analytics present patterns as causes.<br><br>Don't treat patterns as causes unless you're sure.                                                                                                         | Flag PRs that describe correlational analytics as causes or explanations in dashboards, reports, summaries, chart labels, or help text.<br><br>Text says "caused by," "leads to," or "because of" without proof<br>Charts or reports imply reasons instead of just patterns<br>No disclaimers about limits (e.g., "this is only a correlation")<br>Metrics or summaries are described as explanations, not associations<br>Generated insights overstate certainty                                                                                                       | Code level |                                                                                       |                                                                                                                                                                                                    |
| Mismatched Prediction Validation and Use                             | Student-success prediction implementations keep the deployed model consistent with the population, outcome label, and decision use they were validated for. In educational technology, using a model outside its validated scope can trigger inappropriate interventions, miss students who need support, or drive decisions from unreliable signals. Reviewable failures include training or evaluation code that uses a different student population than deployment, label definitions in code, schema, or feature pipelines that do not match the intended outcome, and prediction-serving logic, thresholds, or UI actions that allow use beyond documented constraints or required safeguards.                                                                                                     | Predictions are used beyond what was tested.<br><br>If you use a model outside what it was tested for, you can't trust its results.                                                                     | Flag PRs that deploy or reuse prediction models outside their validated population, outcome, timing, or decision context.<br><br>Training data population differs from deployment population<br>Outcome labels don't match the intended prediction use<br>Prediction logic is used for a different purpose than validated<br>Thresholds or actions don't match validation setup<br>No safeguards to prevent out-of-scope use                                                                                                                                            | Code level |                                                                                       |                                                                                                                                                                                                    |
| Unidirectional Algorithmic Decisions Without Educator Contestability | AI or algorithmic outputs such as risk flags, placement recommendations, or intervention triggers are applied without a structured mechanism for educators, learners, or administrators to submit context, contest a decision, or have that input meaningfully reflected in the system's current or future behavior. In educational technology, this creates a one-way decision flow where human knowledge and contextual judgment cannot correct or qualify what the algorithm surfaces. Reviewable failures include risk or flag workflows that lack a contestation UI, API endpoint, or audit field for human-provided context, systems that accept feedback but do not route it to decision logic or reviewer queues, and missing tests or documentation for override, appeal, or feedback pathways. | Decisions cannot be challenged or corrected by users.<br><br>Decisions go one way, no way for humans to challenge or correct them.                                                                      | Flag PRs that apply risk flags, placements, or interventions without a way for educators or learners to contest decisions, submit context, or trigger review, or where feedback is not routed into decision logic, audit logs, or override workflows.<br><br>No UI/API to submit feedback, appeal, or override decisions<br>No way to add human context to decisions (e.g., notes, comments)<br>Feedback is collected but not used in decision logic or review process<br>No audit trail of human input or changes<br>No tests or docs for override or appeal workflows | Code level |                                                                                       |                                                                                                                                                                                                    |

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| Automation Without Review Safeguards                                 | Consequential learner-affecting actions are not gated by human review or override controls in code. In educational technology, this matters because incorrect automated recommendations, restrictions, or interventions may be applied before a staff member can assess context and stop harm. Failure is visible when UI workflows, API logic, background jobs, permission rules, configuration, or tests let high-impact actions execute automatically, omit an approval step, or lack an authorized path to override or reverse the action.                                                                                                                                                           | High-impact actions skip human review.<br><br>Big decisions should have a human check, not be fully automatic. | Flag PRs that let high-impact learner-affecting actions run automatically without approval, human review, override, reversal, or permission checks.<br><br>Important decisions execute automatically (no approval step)<br>No option to override or cancel decisions<br>No role/permission check for sensitive actions<br>No workflow for human review before applying changes<br>No way to reverse or audit decisions<br>Missing tests for review/approval logic | Code level    |               |                                                                                                                                                                      |
| Missing Consent Checks for Learner Data                              | Learner personal or educational data is processed without enforced consent checks in code. In educational technology, this matters because student data may be collected or shared even when the user never agreed or later withdrew permission. Failure is visible when UI actions, API endpoints, service logic, queues, or tests allow collection, use, or sharing of learner data without verifying current consent state or without blocking processing after consent is revoked.                                                                                                                                                                                                                   | Learner data is used without consent checks.<br><br>Always check permission before using someone's data.       | Flag PRs that collect, use, or share learner data without consent-state checks, revocation handling, access controls, or tests.<br><br>Data is collected, used, or shared without checking consent status<br>No check for revoked or expired consent<br>Processing continues after consent is withdrawn<br>Missing consent fields or validation in APIs/services<br>No tests for consent enforcement                                                              | Code level    | Privacy       | Problems in consent, access, retention, sharing, and protection of learner data. It is about safe and respectful handling of personal and educational information.   |
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| Unenforced High-Risk AI Release Gates                                | High-impact learner-facing AI changes can be released without enforced workflow gates for risk classification, accountable ownership, and approval. In educational technology, this matters because model, prompt, or configuration changes may affect learners before required review occurs. Failure is visible when CI/CD checks, model registry schemas, deployment configuration, feature flags, or tests allow an AI capability to be enabled or updated without required risk tier, owner, and approval fields.                                                                                                                                                                                   | High-risk AI can ship without required checks                                                                  | Flag PRs that enable or update high-risk learner-facing AI without risk tier, accountable owner, approvals, or deployment gates.                                                                                                                                                                                                                                                                                                                                  | Process level | AI Governance | Problems in review gates, release control, monitoring, and ongoing checks for AI-related changes. It is about managing AI risk across development, release, and use. |
| Missing Post-Deployment Risk Monitoring                              | High-impact learner-facing AI features are released without instrumented post-deployment monitoring and threshold-based alerts. In educational technology, this matters because harmful model behavior may appear only after release as learner populations, data, or usage patterns change. Failure is visible when application code, logging schemas, metrics configuration, scheduled evaluation jobs, alert rules, or tests omit required event logging, outcome metrics, drift checks, or alert thresholds for high-risk AI behavior.                                                                                                                                                               | High-risk AI is not monitored after launch                                                                     | Flag PRs that release high-risk AI without logging, drift or outcome metrics, subgroup monitoring, alert thresholds, or response hooks.                                                                                                                                                                                                                                                                                                                           | Process level |               |                                                                                                                                                                      |
| Missing Fairness Review and Validation for Student-Affecting Changes | Student-affecting changes are not gated by required fairness-focused review and targeted validation before merge or release. In educational technology, logic, rule, or configuration updates can change access, support, placement, grading, or opportunity for specific learner groups without anyone checking likely impacts or edge cases. Failure is visible when PR workflows, required fields, reviewer checklists, approval rules, CI checks, or required tests allow changes through without documenting affected learners, impacted decision points, likely downstream educational effects, fairness-specific review questions, targeted edge-case coverage, and resolution of flagged issues. | Student-affecting changes skip fairness review                                                                 | Flag PRs that change learner-affecting logic without documented affected groups, fairness review questions, targeted validation, or edge-case tests.                                                                                                                                                                                                                                                                                                              | Process level |               |                                                                                                                                                                      |
| Missing Bias Review for AI-Assisted Code                             | AI-assisted student-affecting code changes are not gated by required disclosure, human review, and bias-focused validation before merge. In educational technology, coding assistants or generated code can introduce stereotypes, discriminatory logic, or unsafe learner-facing behavior into labels, conditions, prompts, validation rules, or data handling. Failure is visible when PR workflows, source-disclosure fields, reviewer checklists, CI checks, or tests allow AI-assisted or AI-generated changes to merge without being identified and reviewed for bias-related risks.                                                                                                               | AI-assisted code is not checked for bias                                                                       | Flag PRs that include AI-assisted code without disclosure, human review, and bias-focused validation of labels, rules, prompts, or data handling.                                                                                                                                                                                                                                                                                                                 | Process level |               |                                                                                                                                                                      |
| Missing Bias Regression Checks for AI-Generated Code                 | AI code-generation workflows used for student-affecting software are not gated by repeatable bias benchmarks and regression tests before tool, prompt, or model changes are adopted. In educational technology, this matters because discriminatory generated patterns may persist or worsen across versions without consistent measurement. Failure is visible when benchmark prompt sets, scoring scripts, evaluation harnesses, CI checks, or release rules do not quantify and compare bias-related risks in generated labels, conditions, thresholds, proxies, or data-handling logic.                                                                                                              | AI-generated code is not retested for bias                                                                     | Flag PRs that change AI coding tools, prompts, or models without benchmark prompts, scoring scripts, baseline comparisons, or bias regression checks.                                                                                                                                                                                                                                                                                                             | Process level |               |                                                                                                                                                                      |
| Missing User-Need Traceability                                       | User-affecting workflows, defaults, and constraints are changed without visible traceability to documented learner, educator, or administrator needs. In educational technology, this allows technically efficient designs to override autonomy, trust, or practical usability in core learning tasks. Reviewable failures include PRs that add or change required steps, defaults, permissions, nudges, or restrictions without linked requirements, referenced user research findings, acceptance criteria, or tests showing whose need the change serves and how that need is preserved in the implementation.                                                                                        | Changes are not tied to user needs                                                                             | Flag PRs that alter workflows, defaults, permissions, nudges, or restrictions without linked user needs, acceptance criteria, or preserving tests.                                                                                                                                                                                                                                                                                                                | Process level |               |                                                                                                                                                                      |
| Untracked Fairness Mitigation Outcomes                               | In educational systems, a mitigation that is implemented but never tracked can quietly fail while the team assumes the problem is solved. This criterion flags cases where a detected disparity has no assigned owner, no documented rationale for the chosen fix, no post-release evaluation plan, and no escalation path if the fix does not improve group outcomes. Fairness mitigations that are not owned, planned, and validated after release cannot be trusted to hold.                                                                                                                                                                                                                          | Fairness fixes have no owner, plan, or follow-up to confirm they worked                                        | Flag PRs where a fairness disparity has been identified but is missing an assigned owner, documented mitigation rationale, post-release evaluation plan, or escalation path if group outcomes do not improve.                                                                                                                                                                                                                                                     | Process level |               |                                                                                                                                                                      |